

Cloud Functions Deployment

Cloud Functions — Deployment Guide

Source: `/functions/index.js` (Node.js 24, Firebase Functions v2). This guide describes the function inventory, the two-region layout, deployment commands, and common failure modes.

Function inventory

The global default region is `africa-south1` (set in `functions.setGlobalOptions`). Scheduled functions override this and run in `eu-west1` — see the **Regions** section below for why.

africa-south1

Function	Type	Trigger	Purpose
<code>createCustomToken</code>	Callable	HTTPS	Issues a Firebase custom token for clock-number-based login.
<code>sendJobAssignmentNotification</code>	Callable	HTTPS	Pushes an FCM message to a specific recipient token (assignment, busy, dismiss responses).
<code>sendCreatorNotification</code>	Callable	HTTPS	Pushes an FCM message to the operator who created the job.
<code>onJobCardCreated</code>	Firestore trigger	<code>onDocumentCreated('job_cards/{jobId}')</code>	Routes the initial notification to on-site mechanics/electricians per <code>creation_recipients_by_job_id</code> .
<code>onJobCardAssigned</code>	Firestore trigger	<code>onDocumentUpdated('job_cards/{jobId}')</code>	Fires creator/dismiss notifications when <code>assignedClockNos</code> changes.
<code>onCopperTransactionWrite</code>	Firestore trigger	<code>onDocumentWritten('copperTransactions/{docId}')</code>	Notifies copper-authorized users of inventory changes.
<code>onAlertResponseCreated</code>	Firestore trigger	<code>onDocumentCreated('alertResponses/{responseId}')</code>	Handles "I'm Busy" and "Dismissed" responses from notification action buttons.
<code>migrateEmployeeIds</code>	Callable	HTTPS	One-time migration helper.
<code>migrateJobStatuses</code>	Callable	HTTPS	One-time migration helper.

Function	Type	Trigger	Purpose
<code>clearEscalationStamps</code>	Callable	HTTPS	Resets <code>notifiedAtStage1</code> fields when an admin char recipient rules (so old jobs bombard the new audience)

europa-west1

Function	Type	Schedule	Purpose
<code>escalateNotifications</code>	Scheduled	every 2 minutes	Reads <code>notification_configs/global</code> , advances jobs through stages 1→4, fires notifications to the configured recipient rules.
<code>autoCloseMonitoringJobs</code>	Scheduled	<code>0 8 * * *</code> (08:00 daily)	Auto-closes Monitor-status jobs older than 7 days.

Why two regions

Firebase scheduled functions require a region that supports Google App Engine. **africa-south1 does not support App Engine**, so any function using `functions.scheduler.onSchedule({...})` must declare `region: "europa-west1"` (or another App-Engine-supported region). Do not change these to `africa-south1` — deployment will fail.

This split is captured in the memory note `feedback_scheduled_functions_region.md`.

Deployment

From the repo root:

```
# Install / update dependencies
cd functions
npm install

# Deploy everything
firebase deploy --only functions

# Deploy a single function
firebase deploy --only functions:escalateNotifications
firebase deploy --only functions:onJobCardCreated

# Deploy multiple
firebase deploy --only functions:onJobCardCreated,functions:onJobCardAssigned
```

Per-function deploys are strongly preferred during routine work — they're faster and avoid touching the scheduled functions (which take longer to roll over).

The npm scripts in `functions/package.json` shortcut the common commands:

```
npm --prefix functions run deploy # firebase deploy --only functions
npm --prefix functions run logs   # firebase functions:log
```

Required Firestore composite indexes

The escalation function relies on composite indexes for the staged query `where(status == "open").where(notifiedAtStageN == null).where(createdAt <= cutoff)`. The indexes are:

- `status, notifiedAtStage1, createdAt`
- `status, notifiedAtStage2, createdAt`
- `status, notifiedAtStage3, createdAt`
- `status, notifiedAtStage4, createdAt`

These live in `firestore.indexes.json` at the repo root. After modifying:

```
firebase deploy --only firestore:indexes
```

Index quirk: Firestore composite indexes only include documents that actually have the indexed field. Legacy job cards created before the four stamp fields existed won't be returned by these queries until they're backfilled. Use the `clearEscalationStamps` callable to write `null` stamps onto old documents.

Local emulation

```
cd functions
firebase emulators:start --only functions,firestore
```

Override the region in your client when calling the local emulator:

```
FirebaseFunctions.instanceFor(region: 'africa-south1').useFunctionsEmulator('localhost', 5001);
```

Scheduled functions don't fire on a schedule in the emulator — invoke them manually:

```
curl http://localhost:5001/ctp-job-cards/europe-west1/escalateNotifications
```

Common failure modes

Symptom	Likely cause	Fix
<code>escalateNotifications</code> deploy fails with "scheduler not supported in region"	Region was changed to <code>africa-south1</code>	Restore <code>region: "europe-west1"</code> and redeploy
Escalation isn't firing for any job	<code>notification_configs/global</code> is missing, or stages are all disabled	Check the doc exists, and <code>stages.stageN.enabled === true</code> for at least one stage
Escalation isn't firing for a specific job	<code>createdAt <= enabled_at</code> (stage was enabled after the job was created)	Either backfill <code>enabled_at</code> , or run <code>clearEscalationStamps</code> if recipients changed
<code>failed-precondition: query requires an index</code>	Composite index missing	Open the Firebase Console error link to auto-create, then redeploy <code>firestore.indexes.json</code>
Cold-start timeouts on <code>onJobCardCreated</code>	Function instance was idle	Acceptable — first invocation after idle warms up; subsequent calls are fast
Notifications go to the wrong recipients	<code>recipients_by_type</code> keys don't match (mechanical vs mech/elec)	Check <code>jobTypeKey()</code> in <code>index.js</code> — the Dart enum <code>mechanicalElectrical</code> maps to <code>mech/elec</code> (with slash)
Schema drift between admin app and Cloud Function	<code>_buildStageDoc</code> in <code>admin_screen.dart</code> writes a field that <code>getNotificationConfig</code> doesn't read (or vice versa)	Keep both in sync — see <code>escalation_config_structure.md</code>

Logs

```
firebase functions:log
firebase functions:log --only escalateNotifications --lines 200
```

For real-time monitoring, the Firebase Console → Functions → Logs view streams live.

Related docs

- [Escalation system](#) — the algorithm and config schema
- [Troubleshooting](#) — user-facing symptoms that often trace back to functions
- [Firebase security rules](#) — Firestore + Storage access control